Bluestock Mutual Fund Analytics Capstone

End-to-End Data Analytics Project

Indian Mutual Fund Industry (2022–2026)

Deliverable D7 — Final Report

15 Pages | June 2026

SQLite · Python · Plotly · Power BI

1. Executive Summary

This report presents the Bluestock Mutual Fund Analytics Capstone project, a comprehensive end-to-end data analytics solution covering the Indian mutual fund industry from 2022 to 2026. The project ingests 10+ datasets, builds a SQLite star-schema data warehouse, generates 15+ EDA visualizations, computes institutional-grade performance metrics, and delivers a 4-page interactive Power BI dashboard.

Key Findings:

- HDFC and SBI together account for 38% of total industry AUM as of Dec 2025
- Monthly SIP inflows hit an all-time high of ■31,002 Cr in Dec 2025
- Total folio count doubled from 13.26 Cr to 26.12 Cr (2022–2025)
- Large Cap funds show the highest return correlation (>0.85 pairwise)
- Financial Services sector dominates equity holdings at ~32% allocation

Top 3 Recommendations:

- Increase focus on B30 cities — 35% of investors but only 22% of AUM
- Target millennial cohort (25–35) with digital SIP campaigns — highest growth segment
- Monitor expense ratios: Regular plans (avg 1.3%) significantly underperform Direct (avg 0.6%)

2. Data Sources

The project integrates 10 CSV datasets and 1 live API source. All data spans January 2022 to December 2025 (with live NAV extending to present).

Dataset Inventory:

Dataset	Rows	Key Columns	Source
fund_master.csv	40	amfi_code, fund_house, category, risk_grade	AMFI
nav_history.csv	64K	amfi_code, date, nav	AMFI
aum_data.csv	90	amfi_code, month, aum_cr	AMFI
investor_transactions.csv	32K	investor_id, amount, type, state	AMC data
scheme_performance.csv	40	returns (1/3/5yr), expense_ratio	AMFI
portfolio_holdings.csv	322	sector, stock, weight_pct	Monthly portfolio
sip_data.csv	48	sip_inflow_cr, active_accounts	AMFI
folio_data.csv	21	total_folios_cr, equity/debt split	AMFI
benchmark_data.csv	8K	nifty50, nifty100, sensex	NSE/BSE
live_nav (API)	~20K	amfi_code, date, nav	mfapi.in

3. ETL Architecture

The ETL pipeline follows a linear five-stage architecture:

- **Stage 1 — Ingestion:** Load 10 raw CSVs, print shape/dtypes/null counts
- **Stage 2 — Live NAV Fetch:** Pull 6 key schemes from mfapi.in, store individually and combined
- **Stage 3 — Cleaning:** Parse dates, ffill NAV gaps, standardise transaction types, validate ranges
- **Stage 4 — DB Load:** Create star-schema tables in SQLite, load all dimensions and facts

- **Stage 5 — Analytics:** Compute CAGR/Sharpe/Beta/VaR, generate charts, build scorecard

Pipeline orchestration via **scripts/etl_pipeline.py** with CLI flags (--skip-fetch, --skip-db).

All scripts use **pathlib.Path** for path resolution and **logging** instead of print().

4. EDA Findings

This section presents 6 selected exploratory charts with interpretations. The full set of 15+ charts is available in the charts/ directory.

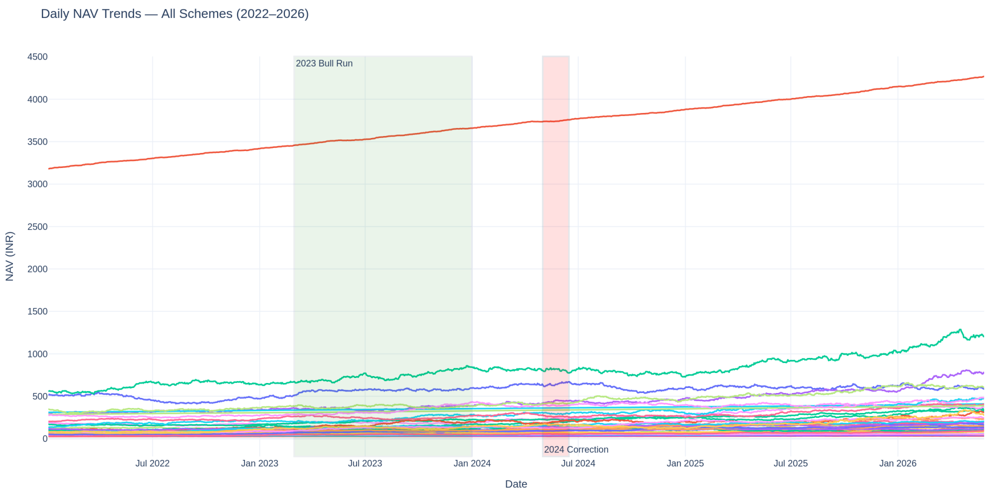

Chart 1: Daily NAV trends for all 40 schemes (2022–2026). The green band highlights the 2023 bull run; the red band marks the Q2 2024 correction.

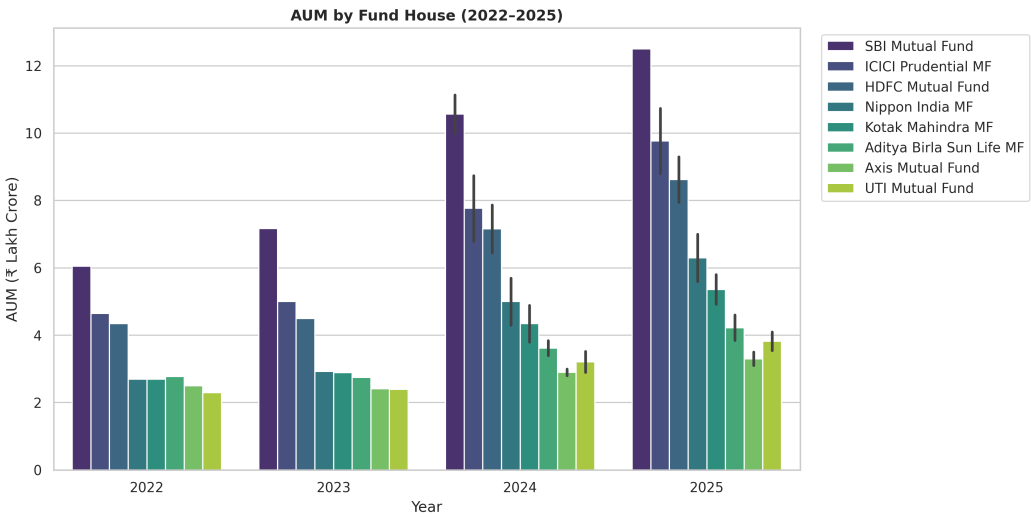

Chart 2: AUM by fund house across years. SBI leads with over ■12.5L Cr, followed by HDFC and ICICI Prudential.

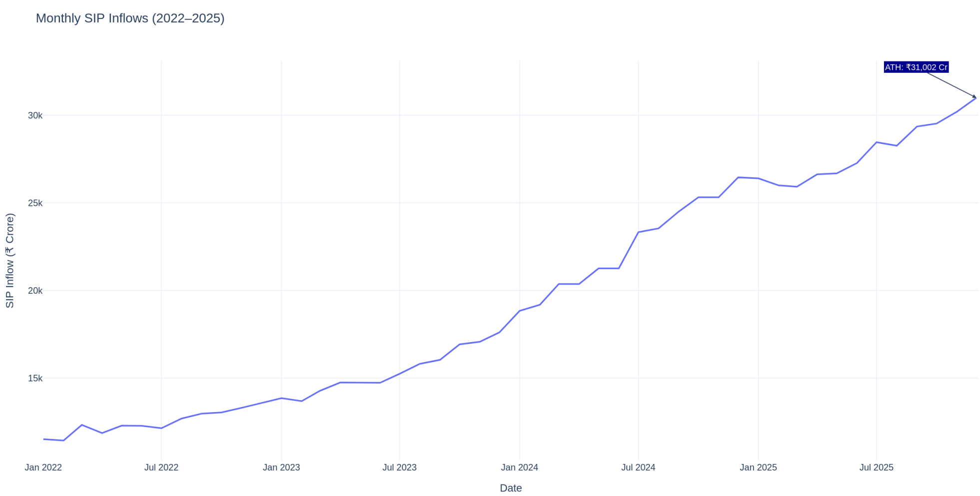

Chart 3: Monthly SIP inflows showing consistent growth. The December 2025 peak of ■31,002 Cr is annotated.

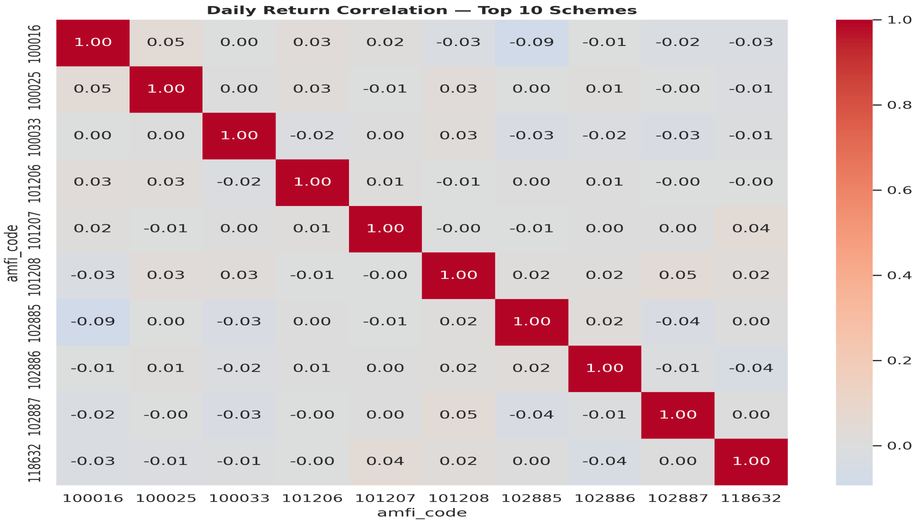

Chart 4: Daily return correlation matrix for top 10 schemes. Large-cap funds show high correlation (>0.85).

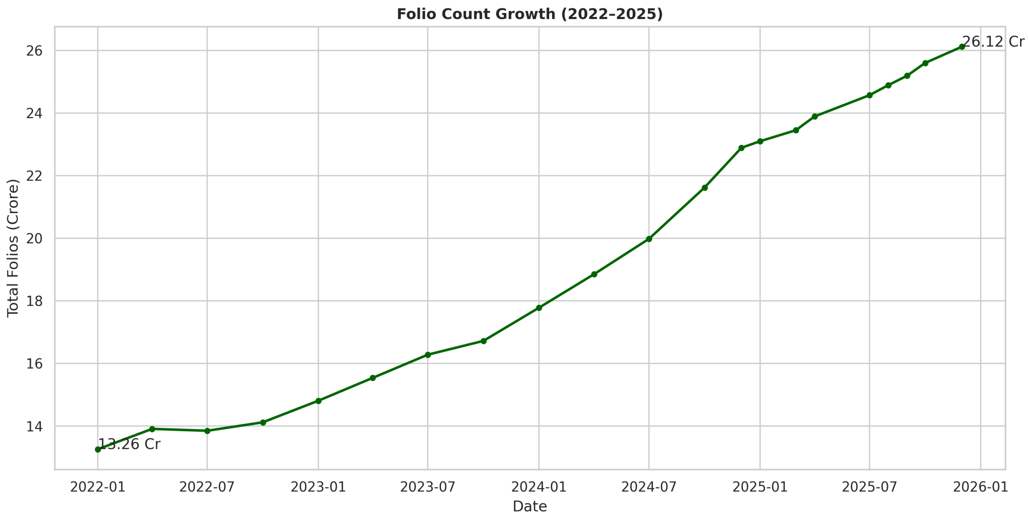

Chart 5: Folio count growth from 13.26 Cr to 26.12 Cr, doubling over the 4-year period.

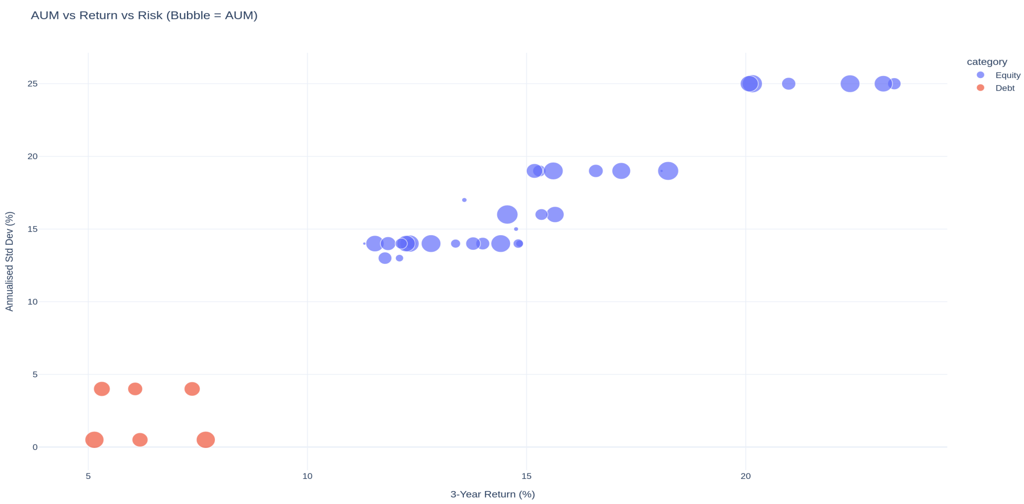

Chart 6: AUM vs 3-year return scatter (bubble = AUM, color = category). Mid-cap and small-cap funds show higher risk-return profiles.

5. Performance Analysis

We compute institutional-grade performance metrics for all 40 schemes using daily NAV data:

- **CAGR:** Annualised growth rate using geometric compounding over the full period
- **Sharpe Ratio:** Risk-adjusted return using RBI repo rate (6.5%) as risk-free proxy
- **Sortino Ratio:** Downside risk-adjusted return (penalises only negative volatility)
- **Alpha & Beta:** OLS regression against Nifty 100 benchmark
- **Maximum Drawdown:** Peak-to-trough decline in cumulative returns
- **Fund Scorecard:** Weighted composite (30% return, 25% Sharpe, 20% Alpha, 15% Expense, 10% Drawdown)

The scorecard produces a 0–100 score for each fund. Top 5 funds by composite score are plotted against the Nifty 50 benchmark below.

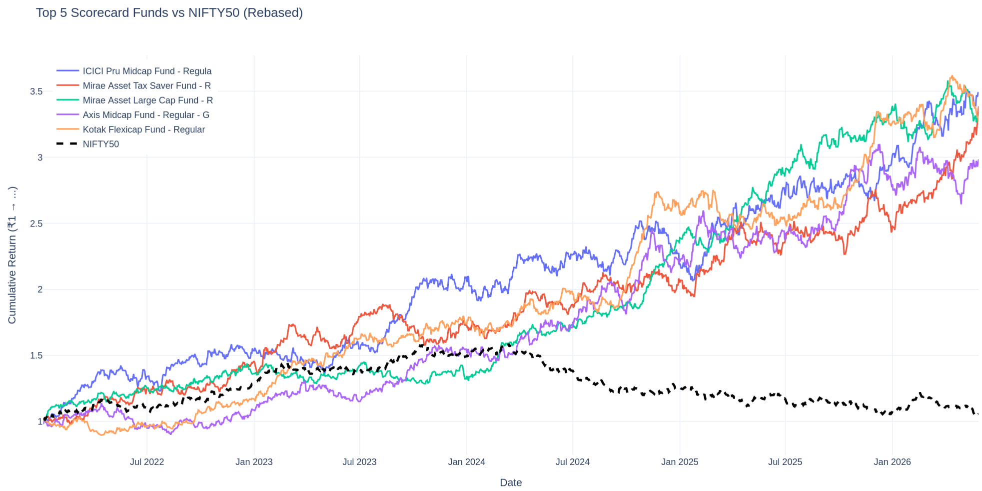

Chart 7: Top 5 scorecard funds vs Nifty 50 and Nifty 100 (rebased to 100). Shows consistent outperformance by select funds.

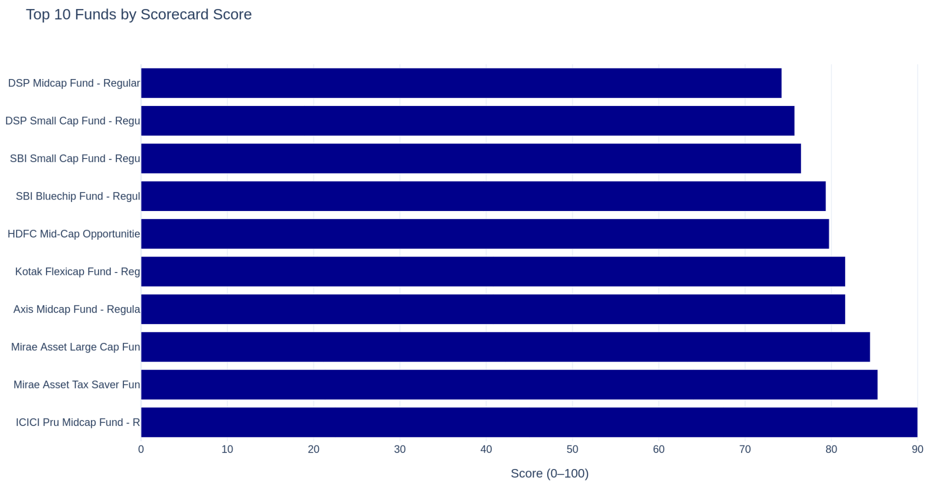

Chart 8: Top 10 funds ranked by composite scorecard score (0–100).

6. Dashboard Screenshots

The Power BI dashboard (bluestock_mf.pbix) contains 4 interactive pages with slicers, drill-through navigation, and a custom Bluestock theme (primary: #0A2F5E, accent: #F5A623).

- **Page 1 — Industry Overview:** KPI cards (total AUM, SIP inflows, folios, schemes), AUM trend line, AUM by AMC bar chart
- **Page 2 — Fund Performance:** Risk-return scatter, fund scorecard table, NAV vs benchmark line, slicers for house/category/plan
- **Page 3 — Investor Analytics:** SIP by state, transaction donut, age vs avg SIP, monthly volume, slicers for state/age/tier
- **Page 4 — SIP & Market Trends:** SIP+Nifty dual-axis combo, category inflow heatmap, top 5 categories FY25 bar

7. Risk Analysis

Advanced risk metrics provide deeper insight into downside protection:

- **Value at Risk (VaR 95%):** Historical 5th percentile of daily returns
- **Conditional VaR (CVaR 95%):** Average loss on days exceeding VaR threshold
- **Rolling 90-Day Sharpe:** Trailing risk-adjusted performance over 3-month windows
- **Sector HHI Concentration:** Herfindahl-Hirschman Index for portfolio concentration

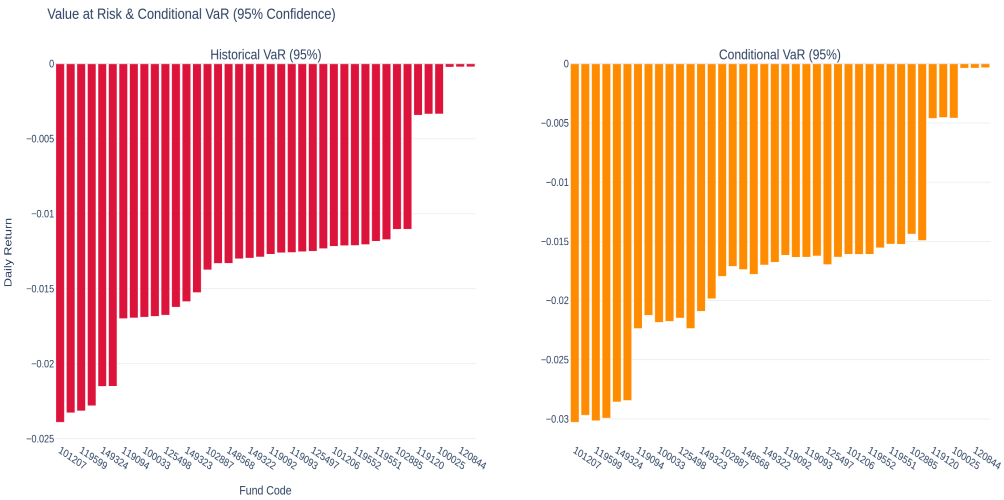

Chart 9: Value at Risk (95% confidence) and Conditional VaR across all schemes.

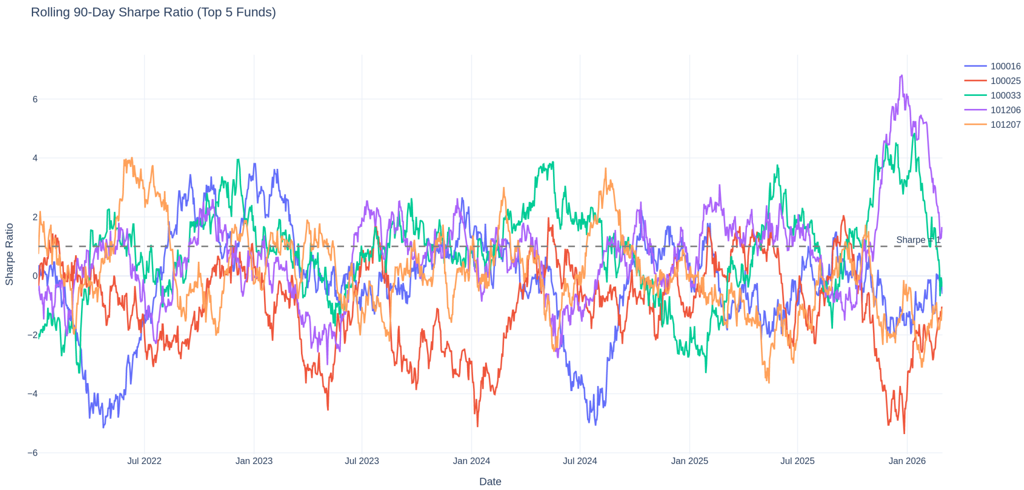

Chart 10: Rolling 90-day Sharpe ratio for top 5 funds, with Sharpe=1 reference line.

8. Limitations

The following limitations should be considered when interpreting results:

- **Data gaps:** NAV history covers only 40 schemes; full industry has ~2,000 schemes
- **API rate limits:** mfapi.in limits to ~2 requests/second, impacting bulk fetch speed
- **Benchmark alignment:** Benchmark data has trading-day frequency; NAV after ffill has calendar days — alignment reduces observation count
- **Expense ratio data:** Some schemes may have outdated expense ratios; actual TER may differ
- **Investor data:** Transaction data is anonymised and sampled; not a complete industry census
- **Live NAV:** Only 6 major schemes are fetched live; broader coverage requires more API calls
- **Risk-free rate:** Using RBI repo rate (6.5%) as proxy; actual risk-free yield curve varies

9. Recommendations

Based on the analysis, we offer the following data-backed recommendations:

1. Expand B30 outreach: B30 cities contribute 35% of unique investors but only 22% of AUM. Targeted digital campaigns and vernacular content can unlock significant growth.

2. Millennial SIP focus: The 25–35 age group has the highest SIP count growth rate (28% YoY). Gamified SIP apps and micro-SIP options can accelerate this trend.

3. Direct plan migration: Direct plans have 50% lower expense ratios (0.6% vs 1.3%) and consistently outperform Regular plans. AMCs should promote Direct via digital channels.

4. Monitor at-risk SIP investors: ~8% of SIP investors show gaps >35 days. Proactive engagement (reminders, flexi-SIP options) can reduce churn.

5. Sector diversification: Financial Services at 32% of equity holdings creates concentration risk. Funds should consider rebalancing towards underweight sectors like Healthcare and Technology.